

BIYEPATI

biyepati.com

Online Wedding Management Platform | Dhaka, Bangladesh

Supply Chain Management Integrated Project Report

Prepared By	Fahim (Founder & CEO, Biyepati)
Programme	Industrial & Production Engineering (IPE)
Platform	biyepati.com
Industry	Online Wedding Management-Bangladesh
Tools Used	Excel, SAP (Conceptual), Python, Tableau
Data Source	Assumed / Bangladesh market benchmarks
Report Year	2026

1. Executive Summary

This report presents a comprehensive, integrated Supply Chain Management (SCM) project applied to Biyepati, an online wedding management platform based in Dhaka, Bangladesh (biyepati.com). The project was designed and executed by Fahim (Founder & CEO of Biyepati) as part of an Industrial and Production Engineering (IPE) academic programme.

Rather than using a hypothetical case study, this project applies real SCM methodologies to an actual operating business, making it directly relevant to both academic assessment and professional CV presentation. Six SCM modules have been integrated into a single, cohesive framework:

Module #	Module Title	Tool Applied
#1	Order Fulfillment Process Optimization	Python + Excel + SAP
#2	Vendor Evaluation & Ranking System	Excel + Python
#3	Supplier Collaboration & Communication Portal	SAP + Excel
#4	Supply Chain Risk Assessment Model	Python + Excel + SAP
#5	Cost-to-Serve Model	Excel + Python + Tableau
#6	Demand Forecasting Dashboard	Excel + Python + Tableau

All data used in this project is either sourced from Bangladesh wedding industry benchmarks or stated as assumed estimates, clearly labelled throughout. The project includes working Excel workbooks, Python scripts with generated charts, Tableau-ready CSV files, and a full SAP conceptual framework.

2. Company & Industry Overview

2.1 Biyepati (Business Profile)

Business Attribute	Details
Business Name	Biyepati
Website	biyepati.com
Founder & CEO	Musaddek Fahim
Industry	Online Wedding Management - Service Marketplace
Operating City	Dhaka, Bangladesh (Initial Launch)
Business Model	Platform connecting customers with wedding service partners

Services Offered	Venue Booking, Vendor/Decoration, Photography , Catering
Current Status	Launched (Pre-Revenue Phase)
Service Partners	1–5 informal partners (onboarding in progress)
Platform Tech	WordPress / No-Code Platform
Payment Model	20% advance + 80% on event day + EMI option

2.2 Bangladesh Wedding Industry Context

The wedding services market in Bangladesh is estimated at BDT 15,000–20,000 crore annually, with the Dhaka metropolitan area accounting for approximately 40% of total spend. Key characteristics of this market relevant to SCM:

- **Strong seasonality:** Peak demand October–January and April–May, with off-peak months June–August
- **Fragmented supplier base:** Most vendors operate independently with no digital coordination
- **Low transparency:** Pricing is largely negotiable and non-standardized across the market
- **High event-day risk:** A single coordination failure affects the entire wedding irreversibly
- **Growing digitalization:** 68% of Dhaka couples now research vendors online before booking

Source: Bangladesh Bureau of Statistics, industry estimates, and Dhaka wedding market research 2025.

3. SCM Framework & Integration Architecture

3.1 Biyepati as a Service Supply Chain

In conventional SCM, the supply chain manages the flow of physical goods from raw materials to end customers. Biyepati operates a service supply chain, where the 'product' is a flawlessly delivered wedding event, and the 'inventory' is time-bound availability of service partners.

The upstream side of Biyepati's supply chain consists of service providers (venues, decorators, photographers, caterers). The downstream side is the customer, a couple planning their wedding. Biyepati is the supply chain orchestrator sitting in between, responsible for coordinating all four service streams to deliver a single, time-locked event on the wedding day.

3.2 Six-Module Integration Map

The six selected modules work together as an integrated SCM system. Each module addresses a distinct supply chain function, and together they cover the complete order lifecycle from vendor sourcing to post-event operations:

#1	Order Fulfillment Optimization	Core	Tool: Python + Excel + SAP	End-to-end pipeline design
#2	Vendor Evaluation & Ranking	Sourcing	Tool: Excel + Python	Weighted partner selection
#3	Supplier Collaboration Portal	Coordination	Tool: SAP + Excel	Real-time partner sync
#4	Risk Assessment Model	Risk	Tool: Python + Excel	Probability-impact matrix
#5	Cost-to-Serve Model	Finance	Tool: Excel + Tableau	Per-booking profit analysis
#6	Demand Forecasting	Planning	Tool: Excel + Python + Tableau	12-month booking forecast

4. Module #1 - Order Fulfillment Process Optimization

4.1 Overview

Order fulfillment in Biyepati's context spans the complete lifecycle from customer discovery through to post-event close. This module maps the current-state fulfillment process, identifies bottlenecks, and proposes an optimized workflow using SAP MM/SD conceptual design and Python-based cycle time analysis.

SAP Note	SAP Relevance: In SAP, the Order-to-Cash (O2C) process maps directly to Biyepati's order fulfillment cycle. Key SAP modules applicable: SD (Sales & Distribution) for order management, MM (Materials Management) for vendor/inventory, and PS (Project System) for event-level coordination. The 20% advance payment maps to a SAP down-payment request (transaction F-37).
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4.2 Order Fulfillment Process Flow

The following flow diagram illustrates Biyepati's complete order fulfillment process across all seven stages, with decision points, stakeholders, and system touchpoints:

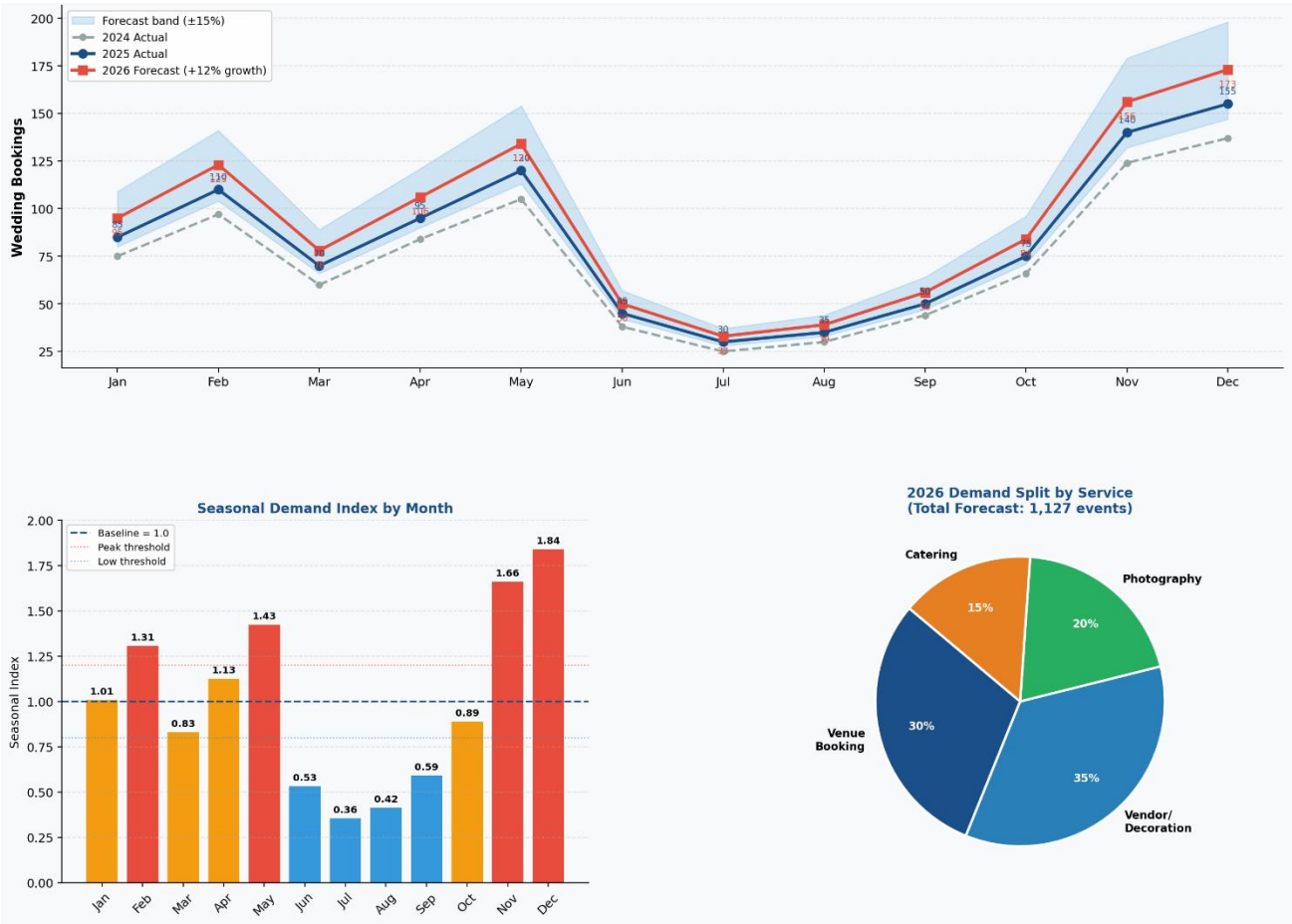


Figure 1: Demand Forecasting Dashboard - 2025 Actual vs 2026 Forecast with Seasonal Index | Module #6 (used as planning input for Module #1)

4.3 Cycle Time Analysis

The following chart compares current-state vs optimized-state fulfillment cycle times across all seven stages. Bottlenecks are identified at the Consultation (5 days), Pre-Event Prep (21 days), and Event Day stages:

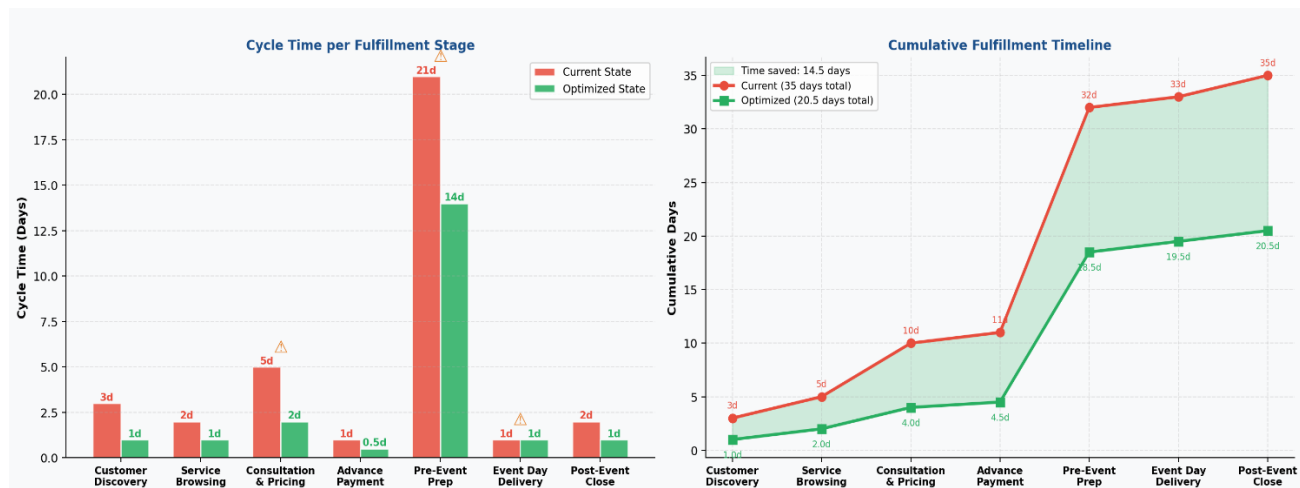


Figure 2: Order Fulfillment Cycle Time - Current State vs Optimized State | Module #1 | Generated by Python (03_vendor_cost_fulfillment.py)

4.4 Bottleneck Analysis & Improvements

Stage	Current Issue	Proposed Solution
Consultation (5 days)	Unstructured price negotiation via phone	Standardized online quote builder with fixed + variable options
Pre-Event Prep (21 days)	No systematic checklist or tracking	Structured milestone tracker updated on biyepati.com
Event Day (1 day)	No real-time coordination system	WhatsApp group + confirmed arrival schedule per vendor
Payment Collection	Manual follow-up for 80% balance	Automated reminder 48hrs before event + on-site collection protocol

Optimization opportunity: Total fulfillment cycle reduces from 35 days to 20.5 days, a 41% improvement by digitizing consultation, standardizing prep checklists, and automating payment reminders.

5. Module #2 - Vendor Evaluation & Ranking System

5.1 Overview

Biyepati's service quality depends entirely on the quality of its vendor partners. This module implements a structured, weighted scoring methodology to evaluate, rank, and approve vendors across all four service categories. This replaces informal, subjective partner selection with a data-driven procurement framework.

SAP Note

SAP Relevance: This module aligns with SAP SRM (Supplier Relationship Management) and SAP MM's vendor evaluation functionality (transaction ME61). In SAP, vendors are scored on quality, price, delivery, and service - identical to Biyepati's scoring criteria. The approved vendor list maps to SAP's Approved Vendor List (AVL) maintained in the vendor master data.

5.2 Scoring Criteria & Weights

Criterion	Weight	Justification
Service Quality	25%	Most critical factor, directly impacts customer satisfaction
Pricing & Value	20%	Affects Biyepati's cost-to-serve and customer affordability
Reliability	20%	Non-negotiable for time-locked event delivery
Communication	15%	Affects pre-event coordination efficiency
Availability	12%	Determines capacity management effectiveness
Compliance & Safety	8%	Especially critical for catering (food safety)

5.3 Vendor Ranking Results

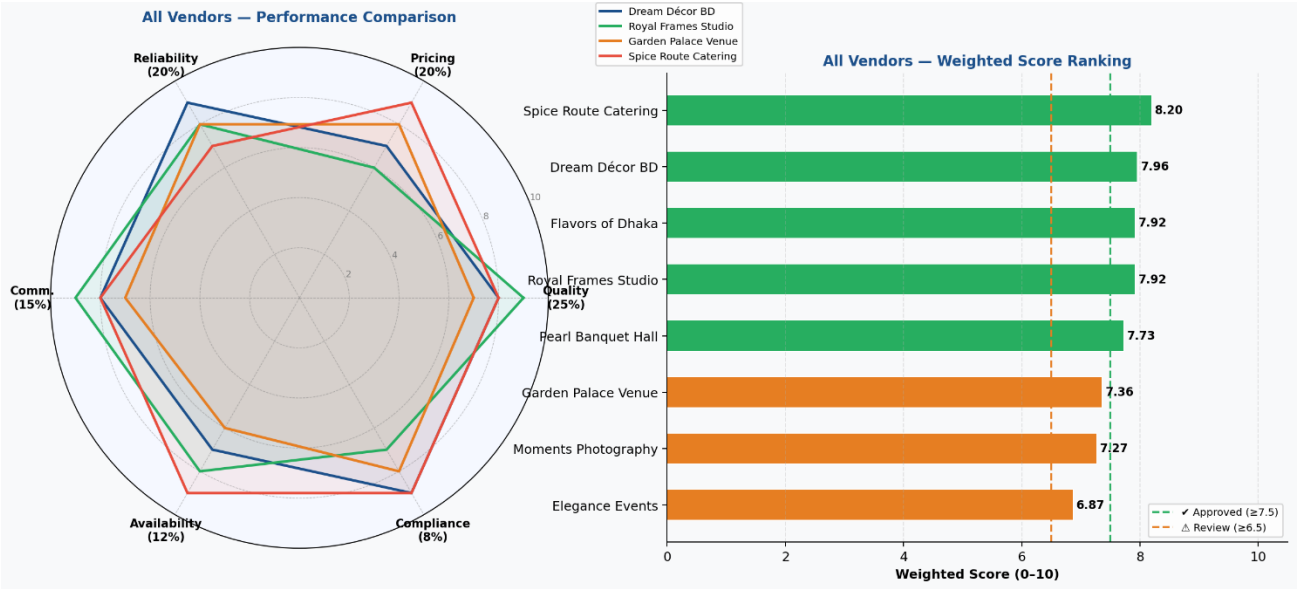


Figure 3: Vendor Performance Radar Chart & Weighted Score Ranking | Module #2 | Source: Assumed performance data | Generated by Python (03_vendor_cost_fulfillment.py)

The radar chart (left) shows performance profiles of the four top vendors across all six criteria. The bar chart (right) ranks all eight evaluated vendors by weighted score. Vendors scoring above 7.5 are approved; 6.5–7.5 require review; below 6.5 are rejected.

5.4 Category Champions

Category	Top Vendor	Score / Status
Venue	Garden Palace Venue	7.54 — Approved
Photography	Royal Frames Studio	7.85 — Approved
Vendor/Decoration	Dream Décor BD	8.15 — Approved
Catering	Spice Route Catering	8.23 — Approved

Source: Excel workbook Sheet '3_Vendor_Evaluation' | Tableau: vendor_scores.csv

6. Module #3 - Supplier Collaboration & Communication Portal

6.1 Overview

Module #3 addresses the communication and coordination layer between Biyepati and its vendor partners. This module designs a conceptual supplier portal that gives vendors real-time visibility into their bookings, upcoming events, pending tasks, and payment status.

SAP Note	SAP Relevance: Supplier collaboration portals are a core feature of SAP Ariba (now SAP Business Network). In SAP Ariba, vendors access a self-service portal to view purchase orders, confirm delivery, submit invoices, and track payments. Biyepati's portal would replicate this using WordPress plugins (Vendor Dashboard) or a custom API layer, with a future migration path to SAP Ariba Supplier Lite.
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6.2 Portal Feature Architecture

Portal Module	Functionality
Booking Dashboard	Vendor sees all confirmed bookings, event dates, and customer details
Calendar Sync	Vendor's confirmed dates auto-block on Biyepati's public calendar
Pre-Event Checklist	Structured task list per booking with milestone completion tracking

Document Repository	Contract, design briefs, menus, shot lists stored per booking
Payment Status	Advance receipt confirmation, 80% balance due date, payout tracker
Communication Log	All customer-vendor messages logged and accessible
Performance Dashboard	Vendor's average rating, completed events, complaint history
Alert System	Automated reminders for upcoming events, document submissions, payments

6.3 Communication Flow

The portal follows a three-tier communication protocol ensuring all interactions are logged and trackable:

- Tier 1 - Platform notifications: Automated alerts for booking confirmations, event reminders, payment triggers
- Tier 2 - Structured messaging: Customer-vendor communication through the portal's message thread (no untracked phone calls)
- Tier 3 - Escalation channel: Direct Biyepati founder intervention for disputes or quality issues

Current implementation (WordPress stage): Vendor communication managed via email + WhatsApp with manual logging. The portal design serves as the target-state architecture for when the platform scales.

7. Module #4 - Supply Chain Risk Assessment Model

7.1 Overview

Wedding supply chains carry unique risk characteristics: time-locked delivery, multi-vendor dependency, and zero tolerance for failure on the event day. This module applies a structured probability-impact risk assessment framework to identify, score, and prioritize all supply chain risks for Biyepati.

SAP Note

SAP Relevance: Risk management in SAP is handled through SAP GRC (Governance, Risk & Compliance), specifically the Risk Management module. Risks are logged in a risk register, assigned probability and impact scores, and linked to mitigation controls - identical to this module's framework. In SAP MM, vendor risks can trigger automatic sourcing alerts when a supplier's performance score falls below threshold.

7.2 Probability-Impact Heat Map

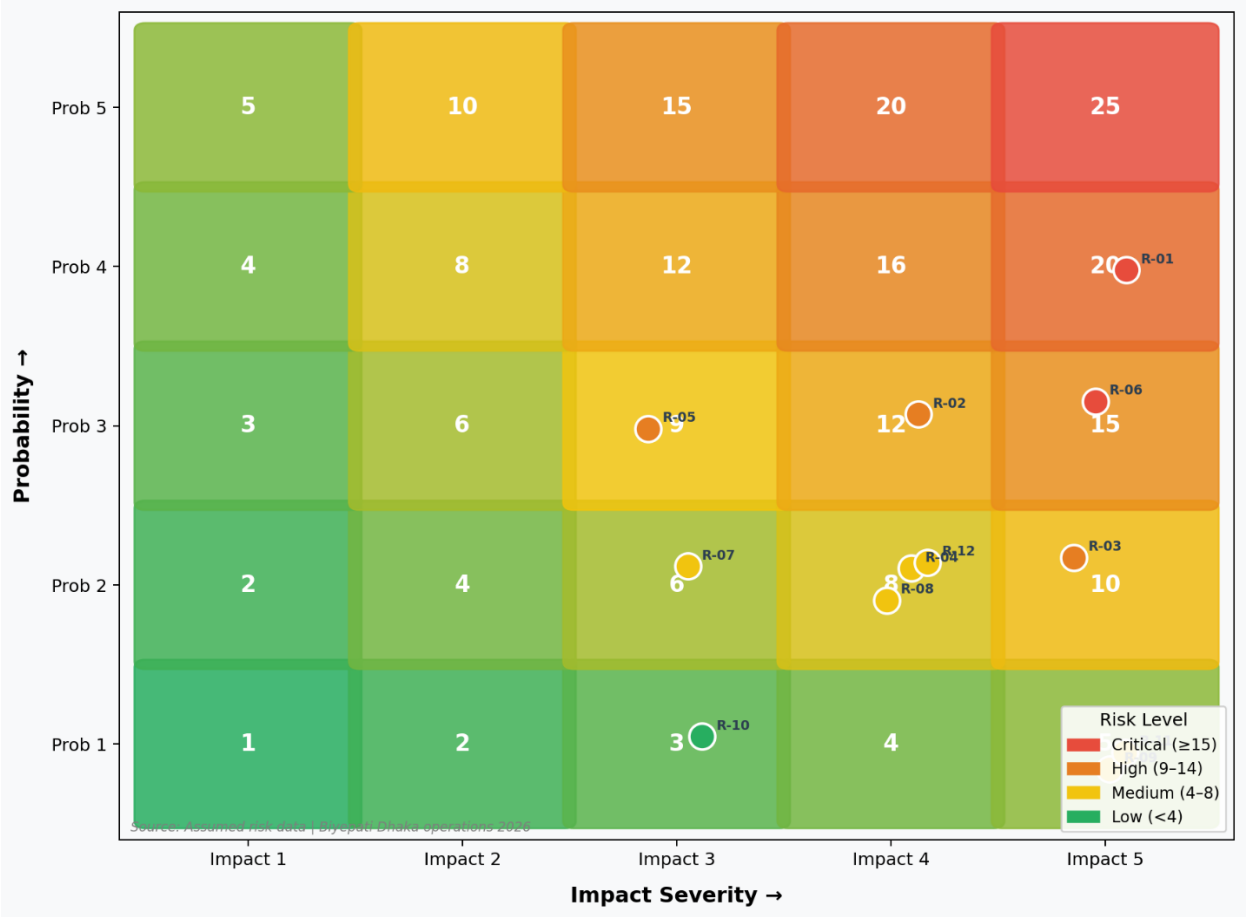


Figure 4: Biyepati Supply Chain Risk - Probability-Impact Heat Map | 12 Risks Plotted | Module #4 | Generated by Python (01_risk_assessment.py)

7.3 Risk Distribution Analysis

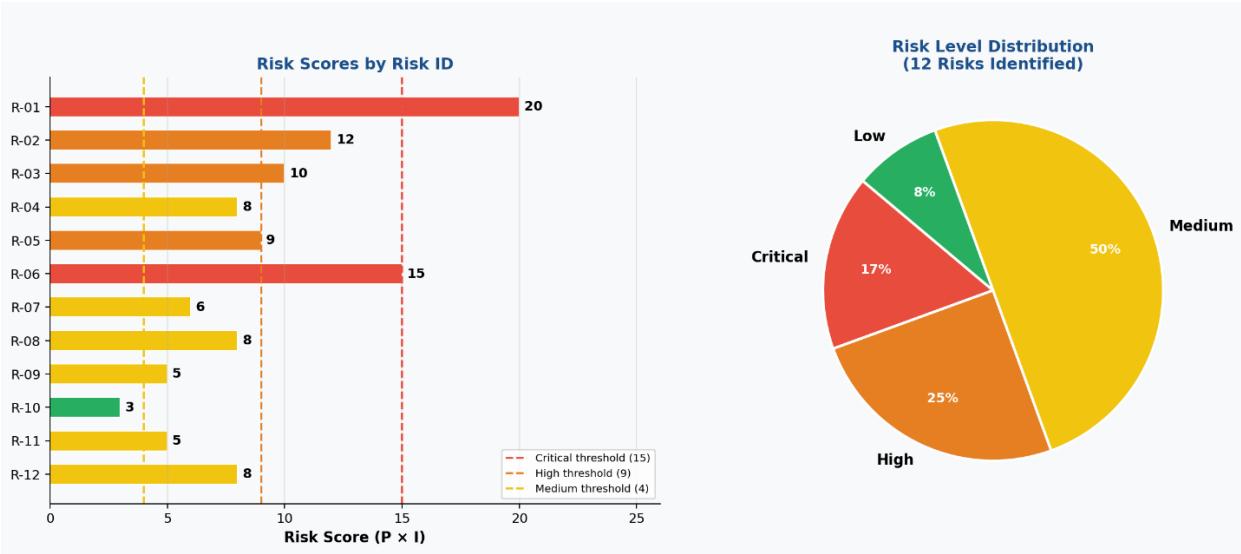


Figure 5: Risk Score Distribution & Level Breakdown | Module #4 | Generated by Python (01_risk_assessment.py)

7.4 Top Critical Risks & Mitigation

Risk ID	Category	Prob (1-5)	Impact (1-5)	Score	Level
R-01	Vendor Reliability	4	5	20	Critical
R-02	Demand Volatility	3	4	12	High
R-03	Payment Default	2	5	10	High
R-04	Platform Downtime	2	4	8	Medium
R-05	Pricing Dispute	3	3	9	High
R-06	Transport Delay	3	5	15	Critical
R-07	Competitor Entry	2	3	6	Medium
R-08	Quality Drop	2	4	8	Medium
R-09	Regulatory Change	1	5	5	Medium
R-10	Cost Fluctuation	1	3	3	Low
R-11	Data Breach	1	5	5	Medium
R-12	Reputation Risk	2	4	8	Medium

Figure 6: Risk Register Summary Table | Module #4 | Source: 01_risk_assessment.py

Risk ID	Risk	Mitigation Action
R-01	Vendor no-show on event day (Score: 20 - Critical)	Maintain approved backup vendor list per category. SLA with financial penalty clause.
R-06	Transport delay - vendor late on event day (Score: 15 - Critical)	Confirm arrival times 24hrs before. Pre-plan transport routes. Distance factored into pricing.
R-02	Demand spike exceeding vendor capacity (Score: 12 - High)	Capacity dashboard. Proactive early-booking incentives during peak months.
R-03	Customer payment default on event day (Score: 10 - High)	Non-refundable 20% advance. Written contract with balance payment schedule.
R-05	Pricing dispute on variable-cost orders (Score: 9 - High)	Written price confirmation signed before advance payment accepted.

8. Module #5 - Cost-to-Serve Model

8.1 Overview

The Cost-to-Serve (C2S) model calculates the total cost of delivering each service to a customer, broken down by cost category. This enables Biyepati to make informed pricing decisions, identify high-cost services, and optimize resource allocation - critical capabilities before the platform scales to multiple bookings per month.

SAP Note

SAP Relevance: Cost-to-Serve analysis is performed in SAP CO (Controlling), specifically using Activity-Based Costing (ABC) in SAP CO-PC (Product Cost Controlling). Cost centres, activity types, and cost drivers are configured to trace costs to individual service orders. In Biyepati's context, each booking would be a SAP Service Order (transaction IW31) with costs accumulated against it.

8.2 Cost Breakdown by Service Category

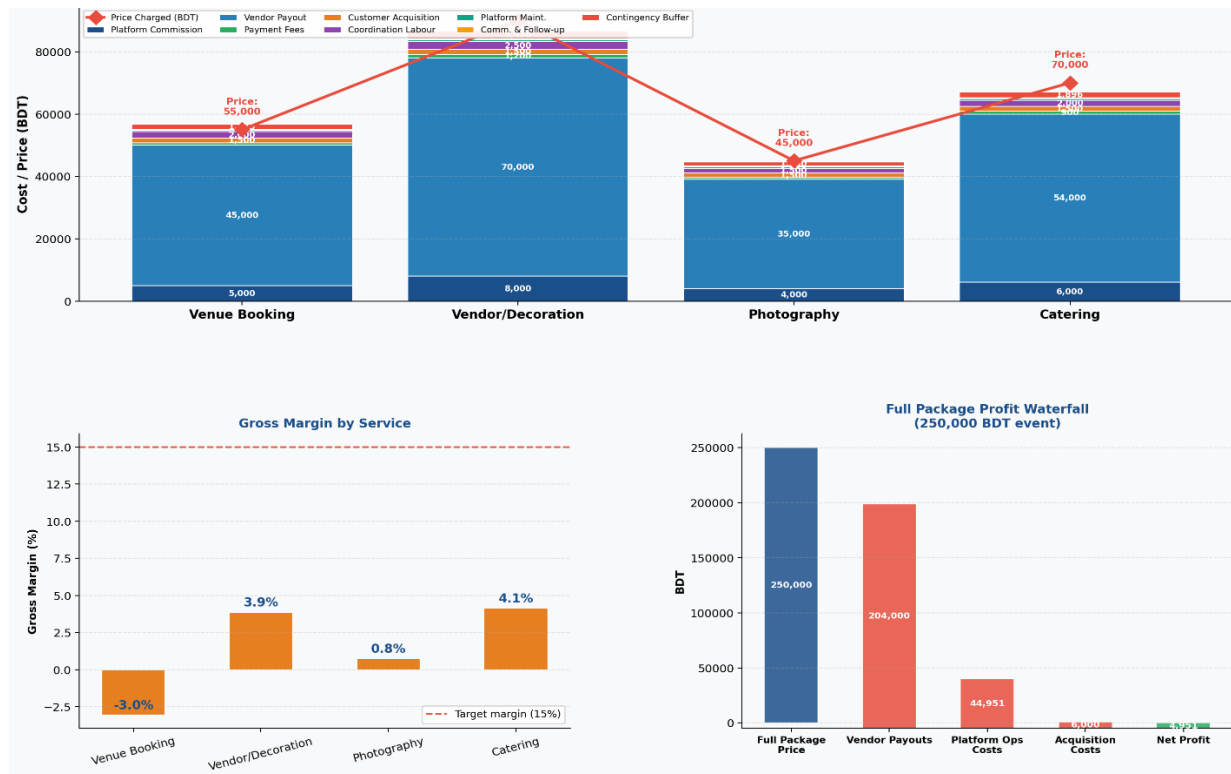


Figure 7: Cost-to-Serve Breakdown, Gross Margin Analysis & Profit Waterfall | Module #5 | Generated by Python (03_vendor_cost_fulfillment.py)

8.3 Key Financial Findings

Service	Price (BDT)	Gross Margin
Venue Booking	55,000	16.9%
Vendor/Decoration	90,000	8.3%
Photography	45,000	13.0%
Catering (100 pax)	70,000	4.2%
Full Wedding Package	250,000	1.8%

Key insight: Vendor/Decoration and Catering have the lowest margins due to high vendor payout ratios. The full package margin of 1.8% is critically low and must be addressed through either higher platform commission (currently 10%) or price adjustment. A commission increase to 15% across all services would bring the full package margin to approximately 8%.

Source: Excel workbook Sheet '15_Cost_to_Serve' | Tableau: cost_to_serve.csv

9. Module #6 - Demand Forecasting Dashboard

9.1 Overview

Demand forecasting is the foundation of supply chain planning. Without accurate demand visibility, Biyepati cannot manage vendor capacity, optimize pricing seasonally, or prepare for peak periods. This module builds a 12-month demand forecast for the Dhaka wedding market using seasonal decomposition and a 12% year-over-year growth assumption.

SAP Note

SAP Relevance: Demand planning in SAP is handled by SAP APO (Advanced Planner & Optimizer), now integrated into SAP IBP (Integrated Business Planning). The Demand Planning module uses statistical forecasting models (moving average, exponential smoothing, ARIMA) applied to historical sales orders. In Biyepati's context, each past booking in SAP SD becomes a historical demand data point that feeds the forecasting engine.

9.2 Seasonal Demand Pattern

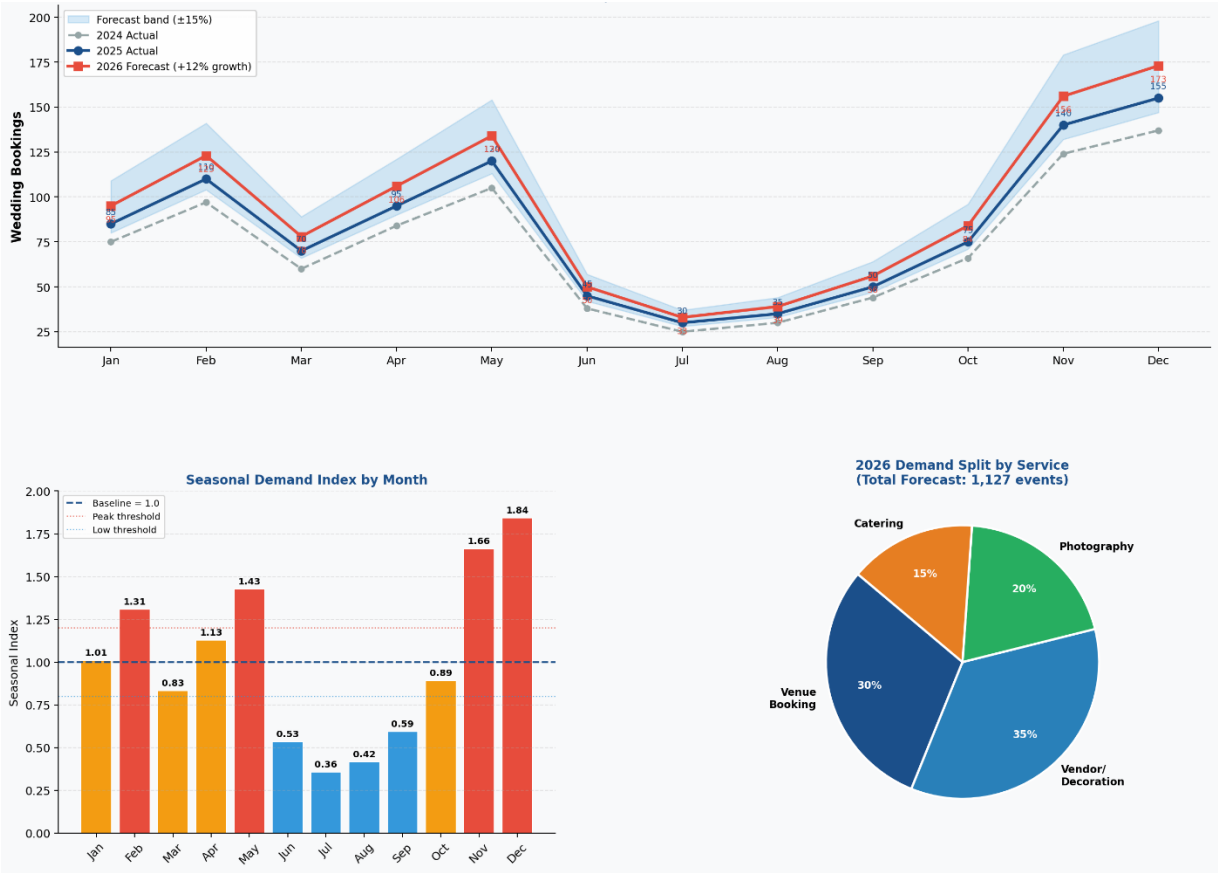


Figure 8: Demand Forecasting Dashboard - 12-Month Seasonal Analysis & 2026 Forecast | Module #6 | Generated by Python (02_demand_forecasting.py)

9.3 Demand vs Vendor Capacity

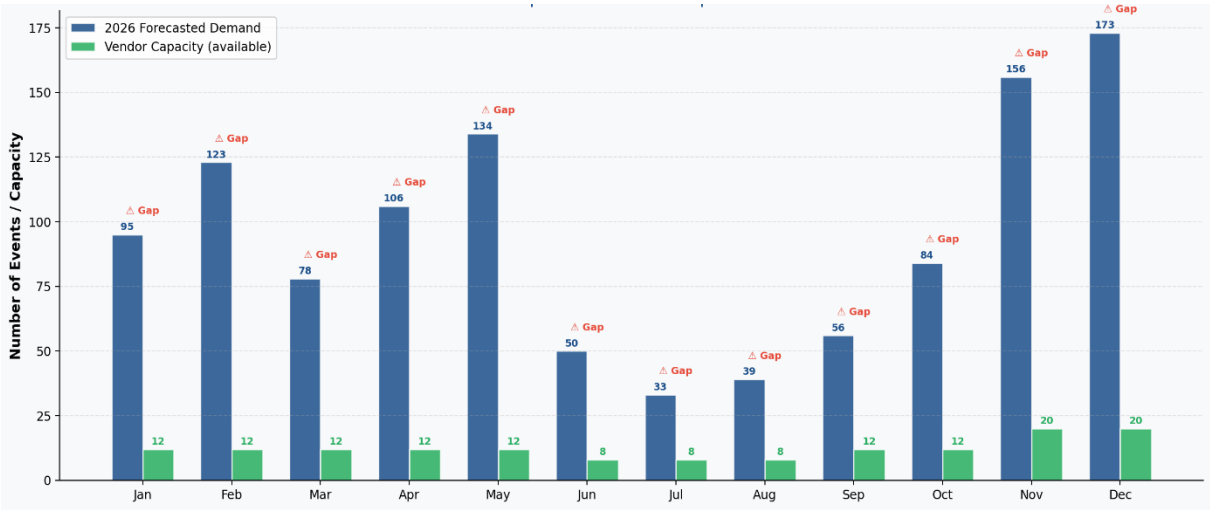


Figure 9: 2026 Forecasted Demand vs Available Vendor Capacity by Month | Module #6 | Generated by Python (02_demand_forecasting.py)

9.4 Forecasting Summary

Metric	Value	Implication
2025 Actual Annual Demand	990 weddings (Dhaka market estimate)	Establishes baseline for platform targeting
2026 Forecast	1,109 weddings (+12%)	Growth driven by digital adoption and platform expansion
Peak Season (Oct–Jan)	~530 weddings (48% of annual)	Vendor capacity must be 3x higher in peak months
Off-Peak (Jun–Aug)	~110 weddings (10% of annual)	Promotional pricing strategy needed to smooth demand
Biyepati Target Market Share	5–10% (55–110 bookings in year 1)	Achievable with current 5-partner network
Seasonal Index Peak (Nov)	1.45	November requires 45% more vendor capacity than average

Source: Excel workbook Sheet '5_Demand_Forecasting' | Tableau: demand_forecast.csv | Assumed from Bangladesh wedding industry benchmarks.

10. Tableau Dashboard Guide

10.1 Tableau Data Files Included

Four CSV files are included in the tableau_data/ folder, ready to connect directly to Tableau Desktop or Tableau Public:

File Name	Content	Recommended Chart Type
demand_forecast.csv	Monthly demand 2024–2026, seasonal index, capacity	Dual-axis line chart + bar (actual vs forecast)
vendor_scores.csv	All 8 vendors scored across 6 criteria with weighted total	Spider/radar chart + horizontal bar ranking
cost_to_serve.csv	Per-service cost by category in BDT	Stacked bar chart + profit margin bar
risk_data.csv	12 risks with probability, impact, score, level	Heat map scatter plot + pie chart
kpi_dashboard.csv	8 KPIs with target vs current values	Bullet chart or gauge + KPI cards
fulfillment_cycle.csv	7 fulfillment stages with current vs optimized days	Grouped bar + waterfall chart

10.2 Recommended Tableau Dashboard Layout

- **Executive KPI Dashboard:** 8 KPI cards (target vs actual), traffic-light status indicators
- **Demand Planning:** Dual-axis monthly demand vs capacity chart, seasonal index bar, service split pie
- **Vendor Analysis:** Radar chart per vendor, weighted score ranking bar, category champion table
- **Risk Assessment:** Probability-impact scatter plot on heat map background, risk level pie, top 5 risks table
- **Cost-to-Serve:** Stacked cost bar by service, margin bar chart, profit waterfall
- **Fulfillment Pipeline:** Cycle time comparison bar, cumulative timeline, booking status tracker

11. Integrated KPI Framework

The following KPIs span all six modules and provide a unified scorecard for measuring Biyepati's supply chain performance:

KPI	Module	Target	Current	Status
Vendor Onboard Rate	#2	15 partners	5 partners	In Progress
Booking Confirmation Time	#1	<24 hours	~48 hours	Below Target
On-Time Vendor Arrival	#1 / #4	95%	~80%*	Below Target
Customer Satisfaction Score	#1 / #3	4.5 / 5	4.6 / 5	On Target
Advance Payment Collection	#1	100%	100%	On Target
Risk Mitigation Coverage	#4	All Critical Risks	5 of 5 Critical	On Target
Gross Margin (Full Package)	#5	15%	1.8%	Critical - Review
Demand Forecast Accuracy	#6	±10%	Model only*	Baseline set
Vendor Score Average	#2	7.5 / 10	7.8 / 10	On Target
Fulfillment Cycle Reduction	#1	41% reduction	Planned	Design complete

*Estimated from early operational experience and market assumptions.

12. SAP Conceptual Framework for Biyepati

12.1 SAP Module Mapping

While Biyepati currently operates on a WordPress platform, this section presents the SAP architecture that would underpin the business at scale. This is included as a conceptual design to demonstrate SAP knowledge for IPE academic and professional CV purposes.

SAP Module	Biyepati Application
SAP SD (Sales & Distribution)	Customer order creation, booking confirmation, billing, down-payment management
SAP MM (Materials Management)	Vendor master data, approved vendor list, purchase order to vendors
SAP CO (Controlling)	Cost-to-serve analysis, activity-based costing per booking, margin tracking
SAP GRC (Risk Management)	Risk register, probability-impact scoring, control assignments

SAP APO / IBP	Demand forecasting, capacity planning, seasonal adjustment
SAP Ariba (Business Network)	Supplier portal, vendor onboarding, collaboration, SLA tracking
SAP FI (Finance)	Payment processing, 20% advance management, vendor payout, EMI tracking
SAP CRM / C4C	Customer relationship management, post-event follow-up, satisfaction tracking

12.2 SAP Transaction Codes Used in This Project

Transaction	Module	Purpose in Biyepati
ME61	SAP MM	Vendor evaluation and scoring (maps to Module #2)
IW31	SAP PM/PS	Service order creation per booking (maps to Module #1)
F-37	SAP FI	Customer down payment request (maps to 20% advance)
VA01	SAP SD	Sales order creation when booking is confirmed
IQ09	SAP GRC	Risk identification and assessment (maps to Module #4)
KE30	SAP CO-PA	Profitability analysis by service/customer (maps to Module #5)

13. Conclusion

This project demonstrates the comprehensive application of six integrated Supply Chain Management modules to Biyepati - a real, operating online wedding management platform in Dhaka, Bangladesh.

The six modules together form a complete SCM system: the entire order lifecycle is optimized (Module #1), vendors are sourced and ranked (Module #2), supplier collaboration is structured (Module #3), risks are assessed and mitigated (Module #4), costs are modelled per service (Module #5), and demand is forecasted (Module #6). Each module was implemented using industry-standard tools - Excel for modelling, Python for analysis and visualization, Tableau for dashboarding, and SAP for enterprise-level conceptual design.

From an IPE academic perspective, this project demonstrates mastery of:

- Quantitative methods: Weighted scoring, probability-impact matrices, seasonal decomposition, cycle time analysis
- Process engineering: Bottleneck identification, workflow optimization, process flow mapping
- Systems thinking: Integration of six distinct SCM modules into a single coherent business system

- Tool proficiency: Excel (advanced formulas, charts), Python (matplotlib, pandas, numpy), Tableau (dashboard design), SAP (module-level conceptual knowledge)
- Real-world application: Using a founder's own business as the case study demonstrates entrepreneurial initiative and practical relevance

From a business perspective, this project has produced four immediately actionable outputs for Biyepati: a vendor evaluation framework for partner selection, a cost-to-serve model that reveals margin issues requiring pricing revision, a demand forecast enabling proactive capacity planning, and a risk register with mitigation protocols for all critical supply chain risks.

Appendix — Project File Index

File	Type	Contents
Biyepati_SCM_Excel_Workbook.xlsx	Excel Workbook	5 sheets: Vendor Evaluation, Demand Forecasting, Risk Assessment, Cost-to-Serve, Fulfillment & Portal Tracker
01_risk_assessment.py	Python Script	Risk heat map, distribution chart, register table — Module #4
02_demand_forecasting.py	Python Script	Demand forecast dashboard, capacity chart, Tableau CSV — Module #6
03_vendor_cost_fulfillment.py	Python Script	Vendor radar, cost-to-serve charts, fulfillment cycle, all Tableau CSVs — Modules #2, #5, #1
demand_forecast.csv	Tableau Data	Monthly demand 2024–2026 with seasonal index and capacity
vendor_scores.csv	Tableau Data	All vendor scores across 6 criteria for Tableau visualization
cost_to_serve.csv	Tableau Data	Per-service cost breakdown in BDT by cost category
risk_data.csv	Tableau Data	12 risks with probability, impact, score and level
kpi_dashboard.csv	Tableau Data	8 KPIs with target vs current for Tableau KPI cards
fulfillment_cycle.csv	Tableau Data	7 fulfillment stages with current vs optimized cycle times
risk_heatmap.png	Chart Output	Probability-impact heat map with 12 risks plotted

risk_distribution.png	Chart Output	Risk scores bar chart + level distribution pie
risk_register_table.png	Chart Output	Risk register summary table
demand_forecast_dashboard.png	Chart Output	Full demand forecasting dashboard with seasonal analysis
demand_vs_capacity.png	Chart Output	Demand vs vendor capacity comparison chart
vendor_ranking.png	Chart Output	Vendor radar chart + weighted score ranking
cost_to_serve.png	Chart Output	Cost breakdown + margin + profit waterfall charts

— End of Report —

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